SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	S.MYTHILI	Reg. No.:	192411250
Section / Batch:		Date:	27/07/2026

Instructions to Students

- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1 Q2,Q3	40 60 (2X30)
	GRAND TOTAL:	100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1. Selection Sort – Dataset with Large and Small Values

Apply the Brute Force technique using Selection Sort on the dataset [99, 4, 76, 15, 2, 61]. Clearly interpret the problem and justify why Selection Sort can be applied effectively to this dataset. Perform the sorting step-by-step and show the intermediate array after each pass. Count the number of comparisons and swaps made during execution. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm.

Q2. Selection Sort – Random Integer Arrangement

Apply the Brute Force technique using Selection Sort on the dataset [38, 12, 47, 5, 29, 18]. Clearly interpret the problem and justify why Selection Sort is suitable for this unsorted dataset. Perform the sorting step-by-step and show the intermediate array after each pass. Count the total number of comparisons and swaps made during execution. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm.

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solutio	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solutio
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%	Application of Concepts (30	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (40 Marks)						
Q2 (30 Marks)						
Q3 (30 Marks)						
				Overall Total (100)		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _______________	Signature: _______________	Signature: _______________
Date: ___________________	Date: ___________________	Date: ___________________

CSA0614 - Design and Analysis of Algorithms :

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192411250

1. **Selection Sort : Dataset With large & SMall Values**

Given dataset :

{99, 4, 76, 15, 2, 61}

Selection Sort finds the smallest element in each pass & Places it in its correct Position

Pass 1 :

Smallest element = 2

Swap 2 & 99

{2, 4, 76, 15, 99, 61}

Pass 2 :

Smallest element = 4

NO Swap required

{2, 4, 76, 15, 99, 61}

Pass 3 :

Smallest element = 61

Swap 61 & 76

{2, 4, 15, 76, 99, 61}

Pass 4:

Smallest element = 61

Swap 61 & 76

{2, 4, 15, 61, 99, 76}

Pass 5:

Smallest element = 76

Swap 76 & 99

{2, 4, 15, 61, 76, 99}

final sorted reply:

{2, 4, 15, 61, 76, 99}

Analysis

Total comparisons → 5 + 4 + 3 + 2 + 1
= 15

total swaps = 4

Time complexity:

Best case = $O(n^2)$

Average case = $O(n^2)$

Worst case = $O(n^2)$

Space complexity = $O(1)$

2. Selection Sort : Random integer Arrangement

Given dataset :

{38, 12, 47, 5, 29, 18}

Selection sort repeatedly finds the smallest element from the unsorted Pair & Places it at the beginning

Pass 1 :

Smallest element = 5

Swap 5 & 38

{5, 12, 47, 38, 29, 18}

Pass 2 :

Smallest element = 12

no swap required

{5, 12, 47, 38, 29, 18}

Pass 3 :

Smallest element = 18

Swap 18 & 47

{5, 12, 18, 38, 29, 47}

Pass 4:

Smallest element = 29

Swap 29 & 38

{5, 12, 18, 29, 38, 47}

Pass 5:

Smallest element = 38

No swap required

{5, 12, 18, 29, 38, 47}

Final sorted array → {5, 12, 18, 29, 38, 47}

Time Complexity:

Best case = $O(n^2)$

Average case = $O(n^2)$

Worst case = $O(n^2)$

Space complexity : $O(1)$

Conclusion: Selection sort successfully sorts the given random dataset. NOT suitable for large datasets because it requires $O(n^2)$ comparisons in all cases.